Contents

Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

Problem 1: Arzela-Ascoli Part I: A bounded sequence of functions on a countable domain has a convergent subsequence

Exercise 6.2.13

Let $A = \{x_1, x_2, x_3, ...\}$ be a countable set. For each $n \in \mathbb{N}$ let f_n be defined on A and assume there exists an $M > 0$ such that $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and $x \in A$.

(a)

Lemma 1: Let $x_1 \in A$ be fixed. Then $(f_n(x_1))$ contains a convergent subsequence.

Proof: Having fixed $x_1 \in A$, the sequence $f_n(x_1)$ is a bounded sequence of real numbers. By the Bolzano-Weierstrass Theorem, it has a convergent subsequence. ■

Consider the subsequence of *functions* generated by taking the functions whose *values* at x_1 give the convergent subsequence above. To indicate that the subsequence of functions f_{n_k} is generated by considering the values of the functions at x_1, we will use the notation $f_{n_k} = f_{1,k}$.

(b)

By a similar argument, the sequence $\left(f_{1,k}(x_2)\right)$ is a bounded sequence of real numbers, so it has a convergent subsequence.

Call this subsequence of functions $\left(f_{2,k}\right)$. Since it is a subsequence of $\left(f_{1,k}\right)$, this sequence of functions, when evaluated at x_1, also converges.

(c)

In this manner, the sequence $f_{n,k}(x_{n+1})$ is a bounded sequence of real numbers, so it has a convergent subsequence. This defines the sequence $\left(f_{n+1,k}\right)$.

$\left(f_{n,k}\right)$ is a subsequence of $f_{m,k}$ whenever $m \leq n$, so $f_{n,k}(x_m)$ converges whenever $m \leq n$.

We will now define a single subsequence $\left(f_{m_k}\right)$ of (f_n):

Let f_{m_1} be the first function in the sequence $\left(f_{1,k}\right)$. Let f_{m_2} be the second function in the sequence $\left(f_{2,k}\right)$. In general, let f_{m_j} be the j'th function in the sequence $\left(f_{j,k}\right)$.

Then, for every j, all but a finite prefix (the first j terms) of $\left(f_{m_k}\right)$ is a subsequence of $f_{j,k}$. It is a valid subsequence since the indices m_k are strictly increasing: $f_{m_{j+1}}$ is the $j+1$'th term of $f_{j+1,k}$, which must appear later in $f_{j,k}$ than f_{m_j} (the j'th term of $f_{j,k}$) since $f_{j+1,k}$ is a subsequence of $f_{j,k}$.

Hence $\left(f_{m_k}(x_i)\right)$ converges for every point $x_i \in A$.

Problem 2: Arzela-Ascoli Part II: Defining equicontinuity

Exercise 6.2.14

A sequence of functions (f_n) defined on a set $E \subseteq \mathbb{R}$ is called *equicontinuous* if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f_n(x) - f_n(y)| < \varepsilon$ for all $n \in \mathbb{N}$ and $|x - y| < \delta$ in E.

(a)

Equicontinuity is a stronger condition than saying that each f_n in the sequence is uniformly continuous, because a single δ works for *every* f_n.

(b)

Let $g_n(x) = x^n$. Individually, each g_n is uniformly continuous on $[0, 1]$ since it's a continuous function on a compact domain. But (g_n) is not equicontinuous. Qualitatively, this is because of what happens close to 1: the sequence gets closer and closer to the limiting function

$$g(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

so the jump from 0 to 1 happens in a shorter and shorter space as n gets bigger. No single δ can suffice for all n, because there will always be some N after which the jump from very-close-to-0 to very-close-to-1 happens in a span of less than δ.

Problem 3: Arzela-Ascoli Part III: A bounded, equicontinuous sequence of functions on a compact domain has a uniformly convergent subsequence

Exercise 6.2.15

For each $n \in \mathbb{N}$, let f_n be a function defined on $[0, 1]$. Suppose that f_n is bounded on $[0, 1]$ – that is, there exists an $M > 0$ such that $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and $x \in [0, 1]$ – and that the collection of functions (f_n) is equicontinuous.

(a)

Let $A = \mathbb{Q} \cap [0, 1]$. Since $\mathbb{Q}$ is countable, A is also countable. By Problem 1, $f_n|_A$ has a convergent subsequence. Equivalently, (f_n) has a subsequence that converges pointwise on every point in A. Call it (g_k).

(b)

Let $\varepsilon > 0$. By equicontinuity, there exists a $\delta > 0$ such that

$$|g_k(x) - g_k(y)| < \frac{\varepsilon}{3}$$

for all $|x - y| < \delta$ and $k \in \mathbb{N}$. Let

$$O = \{V_\delta(r) : r \in A\}$$

be the set of δ-neighborhoods of every rational point in $[0, 1]$. By the density of $\mathbb{Q}$ in $\mathbb{R}$, every real $x \in [0, 1]$ is within δ of some rational point $q \in A$. So O is an open cover of $[0, 1]$. By compactness, it has a finite subcover φ. Let $r_1, r_2, ..., r_m$ be the centers of δ-balls in φ.

The sequence of functions (g_k) from part (a) converges pointwise on every $r \in A$. In particular it converges for every r_i. That is, the sequences

$$(g_k(r_i))$$

for $1 \leq i \leq m$ are convergent sequences of real numbers, hence they are Cauchy. So for each $1 \leq i \leq m$ there exists some $N_i \in \mathbb{N}$ such that for all $s, t \geq N_i$ we have

$$|g_s(r_i) - g_t(r_i)| < \frac{\varepsilon}{3}.$$

Let $N = \max(N_1, N_2, ..., N_m)$, which is defined since m is finite. Then

$$|g_s(r_i) - g_t(r_i)| < \frac{\varepsilon}{3}$$

whenever $s, t \geq N$.

(c)

Let x be an arbitrary point in $[0, 1]$. By construction of the finite subcover φ of δ-balls centered at the rational points r_i, there is at least one r_i such that

$$|x - r_i| < \delta,$$

so

$$|g_k(x) - g_k(r_i)| < \frac{\varepsilon}{3}$$

for all $k \in \mathbb{N}$ by equicontinuity of (f_n) (and therefore of its subsequence (g_k)) and the choice of δ.

Let $s, t \geq N$. Then

$$\begin{aligned}
&|g_s(x) - g_t(x)| \\
&\qquad \leq |g_s(x) - g_s(r_i)| + |g_s(r_i) - g_t(r_i)| + |g_t(r_i) - g_t(x)| \\
&\qquad < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\
&\qquad = \varepsilon.
\end{aligned}$$

Since x was arbitrary and N doesn't depend on x, by the Cauchy Criterion for Uniform Convergence, (g_k) converges uniformly.

Problem 4: A counterexample to show that uniform convergence is not enough for differentiation to commute with limits

Exercise 6.3.3

Define the sequence of functions

$$f_n(x) = \frac{x}{1 + nx^2}.$$

Informally, each f_n is bounded since $\lim_{x\to\infty} f_n(x) = 0$ and $\lim_{x\to-\infty} f_n(x) = 0$ and the denominator is never 0. So outside some compact domain the function goes to 0, and inside the compact domain containing 0 it doesn't blow up. Since it's bounded, its extrema must occur in some compact domain, so they must happen when its derivative is zero. In other words, there must exist some $M > 0$ and $x_0 > 0$ such that $|f_n(x)| < f_n(x_0)$ when $|x| > M$, so we may apply the Extreme Value Theorem to $[-M, M]$. Using the rules of differentiation, that happens when

$$x = \pm\frac{1}{\sqrt{n}}.$$

Here is a proof that $x_0 = 1/\sqrt{n}$ is a global maximum for each f_n. $f_n(x_0) = \frac{1}{2\sqrt{n}}$. Applying the AM-GM inequality to 1 and nx^2, we have

$$\frac{1 + nx^2}{2} \geq \sqrt{n}x.$$

When $x > 0$, rearranging gives

$$\frac{x}{1 + nx^2} \leq \frac{1}{2\sqrt{n}}$$

as desired. This clearly holds for $x = 0$ as well. Since $f_n(x) < 0$ when $x < 0$, this is a global maximum.

Since f_n is an odd function, it has a global mininum at $-x_0$ with value $-\frac{1}{2\sqrt{n}}$.

Thus

$$|f_n(x)| < \frac{1}{2\sqrt{n}}$$

for all $x \in \mathbb{R}$. We use this to show uniform convergence of (f_n) to the constant function $f(x) = 0$.

Let $\varepsilon > 0$ be given. Then pick N large enough such that

$$|f_N(x) - f(x)| = |f_N(x)| < \frac{1}{2\sqrt{N}} < \varepsilon.$$

This is possible since $\frac{1}{2\sqrt{n}}$ converges to 0, and by the Archimedean Property. Since $\frac{1}{2\sqrt{n}}$ is a strictly decreasing function of n, we have

$$|f_n(x)| < \varepsilon \qquad \text{whenever} \qquad n \geq N.$$

Since this choice of N works for all $x \in \mathbb{R}$, the convergence is uniform.

(b)

Let $f = \lim f_n$. Then $f(x) = 0$.

Using the rules of differentiation,

$$f_n'(x) = \frac{1 - nx^2}{(1 + nx^2)^2}$$

so

$$\lim f_n'(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$$

since when $x = 0$ the function is identically 1, and when $x \neq 0$ we may divide the numerator and denominator by n^2 to see that the numerator goes to 0 while the denominator goes to $x^4 \neq 0$, so the limit is 0 by the Algebraic Limit Theorem.

(This already tells us that the sequence of derivatives doesn't converge uniformly: each derivative is continuous on all of $\mathbb{R}$ since it's a rational function with nonzero denominator, but the limit function has a discontinuity at zero.)

So, in this case, $f'(x) = \lim f_n'(x)$ for every $x \in \mathbb{R} \setminus \{0\}$.

Problem 5: If derivatives converge uniformly and the sequence converges at any point, it converges uniformly

Exercise 6.3.7

Proof idea

This is a simpler version of the technique used in the proof of the Differentiable Limit Theorem: apply MVT to $f_n - f_m$ and use the Cauchy Criterion for Uniform Convergence to bound the difference in the derivatives at a point.

My loose, intuitive idea of this theorem is that if the derivatives converge uniformly, then all the functions differ at most by a constant. So if we can pin down that constant, then the functions actually have to agree.

Proof

Theorem 1: Let (f_n) be a sequence of differentiable functions defined on the closed interval $[a, b]$, and assume (f'_n) converges uniformly on $[a, b]$. If there exists a point $x_0 \in [a, b]$ where $f_n(x_0)$ is convergent, then (f_n) converges uniformly on $[a, b]$.

Proof: Let $\varepsilon > 0$ be given and let $x \in [a, b]$ be arbitrary.

Since the sequence of real numbers $(f_n(x_0))$ converges, it is Cauchy, so there exists some $N_1 \in \mathbb{N}$ such that

$$|f_n(x_0) - f_m(x_0)| < \frac{\varepsilon}{2} \tag{1}$$

whenever $m, n \geq N_1$.

By uniform convergence of (f'_n) and the Cauchy Criterion for Uniform Convergence, there exists some $N_2 \in \mathbb{N}$ such that

$$|f'_n(y) - f'_m(y)| < \frac{\varepsilon}{2(b-a)}$$

whenever $m, n \geq N_2$ for all $y \in [a, b]$.

Let $N = \max(N_1, N_2)$ and let $m, n \geq N$.

If $x = x_0$ then

$$|f_n(x) - f_m(x)| = |f_n(x_0) - f_m(x_0)| < \frac{\varepsilon}{2} < \varepsilon$$

by Equation 1.

Otherwise, by the Mean Value Theorem applied to the function $f_m(y) - f_n(y)$ on the interval $[x_0, x] \subseteq [a, b]$ (if $x < x_0$ the argument is the same), there exists some $\alpha \in [x_0, x]$ such that

$$\frac{(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0))}{x - x_0} = f'_n(\alpha) - f'_m(\alpha),$$

but

$$|f'_n(\alpha) - f'_m(\alpha)| < \frac{\varepsilon}{2(b-a)},$$

so

$$|(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0))| < |x - x_0| \frac{\varepsilon}{2(b-a)}.$$

Since $x, x_0 \in [a, b]$,

$$|x - x_0| \leq b - a,$$

so

$$|(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0))| < |x - x_0| \frac{\varepsilon}{2(b-a)} \leq \frac{\varepsilon}{2}.$$

Now by the triangle inequality,

$$\begin{aligned}
&|f_n(x) - f_m(x)| \\
&\qquad \leq |(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0))| + |f_n(x_0) - f_m(x_0)| \\
&\qquad < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&\qquad = \varepsilon.
\end{aligned}$$

So by the Cauchy Criterion for Uniform Convergence, since the choice of N was independent of x, (f_n) converges uniformly. ■